(SCTP) – Full Stack Web Development CAPSTONE PROJECT

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About this module

This module is NOT a typical training module, but more like a **mentoring module**.

You are going to work on given project in each session, and your trainer will guide you wherever you need help.

Trainer will keep you on track if your approach deviates from the project objectives.

Trainers are not supposed to write the codes for you or complete the tasks for you. They may give some suggestions to help you complete the tasks.

Assessment Methods

Practical Performance

Presentation

Important to note:

- Passing criteria: 100% score
- Assessment methods:
  - Practical Performance: The entire duration of Practical Performance entails you work on your respective projects.
  - Presentation: Last 2-3 sessions of the module will be used for the presentation.
  - Individual presentation duration: 15 minutes

Practical Performance (PP)

1. Problem Statement

In today's fast-paced world, if libraries often rely on manual methods to track books, users, and transactions, can lead to misplaced items, delays, and frustration for both staff and visitors. The goal of this project is to build a **user-friendly, web-based system** that automates **key processes like borrowing, returning**, and inventory management.

By reducing paperwork and providing real-time data, librarians can focus on delivering better service and an improved user experience.

What is your MAIN task?

Your task is to develop a **Library Management System** using **Spring Boot**, **MySQL,** and **React**. The system should automate book tracking, member management, and borrowing processes.

It must provide an **easy-to-use interface** for librarians to manage books and members, and for users to search, borrow, and track books. The goal is to create a modern, efficient solution that simplifies library operations and improves the experience for both staff and users.

What are the REQUIREMENTS?

- **Collect and Structure Data**: Gather all relevant library information (books, users, borrow/return records) and ensure it is accurate, consistent, and ready for the system.
- **Define Your Problem Statement and Objectives**: Clearly outline the main challenge and specific goals for your Library Management System (e.g., improved tracking, better user experience).

What are the REQUIREMENTS?

- **Choose Appropriate Tools**: You will use Spring Boot, MySQL, and React for core development.
- **Cover All Project Stages**: Ensure your solution covers planning, design, development, testing, phases addressing every essential step of building a robust system.
- **Present Your Work and Findings**: Prepare a final presentation detailing the system's features, key recommendations or future improvements.

Presentation (P)

What is the EXPECTATION?

You are expected to provide a **clear and concise presentation** of your project to the assessor with the following markers and rubrics.

Duration: Max 15 mins (+ 5 mins for Q&A)

Markers and Rubrics for Project

Presentation Markers	Rubrics	Details (for Library Management System)
1. Problem Statement & Objectives	- Clarity and completeness of the problem description. Relevance and feasibility of objectives	- Clearly state the current issues with manual or outdated library processes - Define key objectives: improve book tracking, enhance user experience, reduce errors, etc.
2. Project Scope & Requirements	- Comprehensive coverage of system needs. - Alignment with problem statement - Realistic scope and requirements	- Outline major functionalities: user registration, catalog management, borrow/return transactions, and reports.

Markers and Rubrics for Project

Presentation Markers	Rubrics	Details (for Library Management System)
3. Technical Stack & System Architecture	- Justification for chosen technologies - Logical and well-organized architecture - Clear explanation of data flow and components	- Present how Spring Boot (back-end), MySQL (database), and React (front-end) interact. - Show an architecture diagram illustrating layers (UI, API, Database).
4. Implementation & Code Quality	- Code organization and readability - Adherence to best practices (RESTful APIs, database design) - Successful integration of all components	- Demonstrate CRUD operations for books, users, and transactions. - Show how your application handles user roles (librarian vs. patron). - Provide examples of well-structured classes, services, and controllers in Spring Boot.

Markers and Rubrics for Project

Presentation Markers	Rubrics	Details (for Library Management System)
5. Testing & Validation	- Coverage of unit, integration, and user acceptance tests - Effectiveness in finding and fixing issues - Demonstration of a stable, error-free system	- Show how JUnit tests validate Spring Boot services and repository layers. - Present integration tests for front-end and back-end (React + Spring Boot). - Include user acceptance testing for critical workflows like borrowing and returning books.

Markers and Rubrics for Project

Presentation Markers	Rubrics	Details (Library Management System Context)
6. Presentation Delivery & Q&A	- Clear, concise communication - Logical flow and engagement - Ability to address audience questions and feedback	- Summarize how the Library Management System meets its objectives. - Use diagrams, screenshots, or short demos to illustrate key features.

Deliverables

- You are expected to put together the following, bundled in a <YourName>**.zip** file and upload it on CANVAS under Practical Performance assessment:
  - Source code – Front-end (React) and Back-end (Spring Boot)
  - Database – ERDs, SQL Script Files for database, table creation, various insert queries etc
  - Documentation – BRD and SRS, System Design Documents
  - Manuals – User manual
- Final Presentation (**.pptx**) to be uploaded separately in the Presentation assessment on CANVAS:
  - Presentation should include all the details as per the Presentation Markers
- Both submissions must be done prior to the presentation start, typically 2-3 sessions in advance. Trainer will confirm on the actual date.

Tentative Timeline

Tentative Milestones

Milestone	Action	Deliverable	Duration in Hrs (approx.)
M1	Project Setup	Basic project setup structure	4 hrs
M2	User Registration & Authentication	Working user registration and login feature with corresponding documentation	6 hrs
M3	Book Management	Working admin dashboard to manage books with design and API docs	8 hrs
M4	Book Search & Borrow/Return	Integrated book search, borrow, and return functionalities with documentation	10 hrs
M5	Overdue Notifications	Overdue notification implementation	4 hrs
M6	Integration, Testing and Final Documentation	A completed project with documentation and presentation	8 hrs

THANK YOU!

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